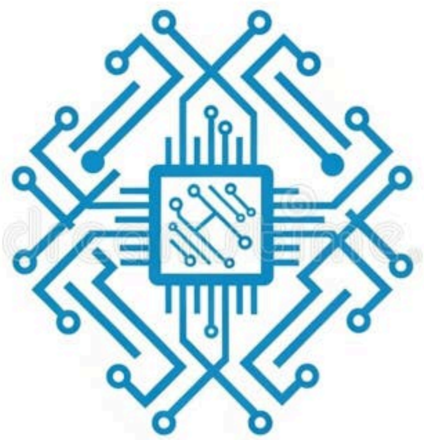
This unified syllabus is based on common patterns across AICTE model curricula and major Indian universities. Exact titles, sequencing, and electives vary by institution; use this as a detailed, chapter-wise master reference when mapping to your university's official syllabus.



Semester 1

Subject	Chapters & Topics (chapter names with key topics)

Engineering Mathematics I (Calculus & Linear Algebra)	1. Limits, Continuity, and Differentiability — epsilon-delta intuition, continuity tests, differentiability, Rolle’s and Mean Value Theorems | 2. Applications of Derivatives — monotonicity, extrema, curvature and concavity, Taylor’s series, error bounds | 3. Integral Calculus and Applications — definite integrals, improper integrals, reduction formulae, areas and volumes of revolution | 4. Multiple Integrals — double and triple integrals, Jacobians, change of order and variables, applications to area and mass | 5. Vector Calculus Basics — gradient, divergence, curl, line and surface integrals, Green’s, Gauss’ and Stokes’ theorems (statements and applications) | 6. Matrices and Linear Algebra Foundations — rank, echelon forms, systems of linear equations, eigenvalues and eigenvectors, diagonalization
Programming for Problem Solving (C Language)	1. Algorithms, Flowcharts, and C Program Structure — problem decomposition, compilation model, header files, main function | 2. Data Types, Operators, and Expressions — integer and floating types, arithmetic and bitwise operators, precedence and associativity | 3. Control Flow and Modular Programming — selection, iteration, nested loops, functions, scope, storage classes | 4. Arrays, Strings, and Parameter Passing — one/two-dimensional arrays, string handling, pass-by-value vs arrays | 5. Pointers and Dynamic Memory Management — pointer arithmetic, pointers with arrays and functions, malloc/calloc/free | 6. Structures, Unions, and File Handling — composite types, typedef, binary/text files, formatted I/O, basic error handling

Engineering Physics (Foundations for Computing)	1. Oscillations and Wave Mechanics — damped/forced oscillations, resonance, wave equation, superposition | 2. Optics and Photonics — interference and diffraction, polarization, lasers, optical fiber principles | 3. Quantum Mechanics Basics — wave-particle duality, Schrödinger equation (concept), particle in a box, tunneling | 4. Semiconductor Physics — energy bands, intrinsic/extrinsic semiconductors, carrier dynamics, PN junction | 5. Magnetic and Dielectric Materials — ferromagnetism, hysteresis, polarization, dielectric breakdown | 6. Nanoscience Applications — quantum confinement, synthesis routes, characterization techniques, device perspectives
Basic Electrical and Electronics Engineering	1. DC Circuit Analysis — Kirchhoff laws, node/mesh methods, Thevenin and Norton equivalents | 2. AC Circuits and Power — phasors, impedance, resonance, power factor correction | 3. Transformers and Rotating Machines (Overview) — ideal transformer, losses, induction motor fundamentals | 4. Diodes and Rectifiers — VI characteristics, half/full-wave rectifiers, filters, Zener regulation | 5. Transistors and Amplifiers — BJT/FET basics, biasing, small-signal model, common amplifier stages | 6. Operational Amplifiers and Applications — ideal op-amp, inverting/non-inverting, active filters, instrumentation basics

Engineering Graphics and CAD	1. Drawing Standards and Scales — conventions, line types, dimensioning norms, scale construction | 2. Orthographic Projections — first-angle/third-angle systems, projection of points, lines, and planes | 3. Projections of Solids and Sections — prisms, pyramids, cylinders, cones, sectional views, true shapes | 4. Isometric and Perspective Projections — isometric scales, isometric views, visual depth representation | 5. Development of Surfaces — sheet metal development, transition pieces, intersection of solids | 6. Computer-Aided Drafting Basics — CAD interface, layers, blocks, constraints, plotting and printing
Professional Communication and Ethics	1. Foundations of Technical Communication — clarity, coherence, audience analysis, tone | 2. Grammar, Vocabulary, and Mechanics — common errors, conciseness, punctuation for impact | 3. Technical Writing Forms — lab reports, project proposals, research abstracts, user manuals | 4. Oral Communication and Presentations — structuring talks, visual aids, delivery and Q&A | 5. Workplace Communication — emails, memos, meeting minutes, etiquette, intercultural aspects | 6. Professional Ethics and Values — responsibility, integrity, whistleblowing, case studies in technology

Semester 2

Subject	Chapters & Topics (chapter names with key topics)

Engineering Mathematics II (Differential Equations & Transforms)	1. First-Order Differential Equations — exact, linear, Bernoulli, applications to growth/decay | 2. Higher-Order Linear ODEs — constant coefficients, undetermined coefficients, variation of parameters | 3. Laplace Transforms and Applications — shifting, convolution, inverse transform, ODE solving | 4. Fourier Series and Transforms — orthogonality, half-range expansions, frequency domain intuition | 5. Partial Differential Equations — formation, Lagrange’s linear PDE, heat and wave equations | 6. Probability and Statistics Primer — random variables, distributions, estimation, confidence intervals
Engineering Chemistry for Computing	1. Chemical Bonding and Molecular Structure — hybridization, VSEPR, intermolecular forces | 2. Water Technology and Corrosion Control — hardness, softening, corrosion mechanisms, protection | 3. Polymers, Composites, and Smart Materials — synthesis, properties, engineering uses | 4. Fuels, Energy, and Combustion — calorific value, alternative fuels, green energy | 5. Electrochemistry and Batteries — Nernst equation, electrodes, primary/secondary cells, fuel cells | 6. Nanomaterials and Green Chemistry — synthesis approaches, catalysis, sustainability principles

Digital Logic Design	1. Number Systems and Boolean Algebra — complements, minimization, Boolean theorems | 2. Logic Gates and Combinational Design — adders, subtractors, multiplexers, decoders | 3. Simplification Techniques — K-maps, Quine-McCluskey, don’t care handling | 4. Sequential Circuits and Latches — SR/JK/T/D, flip-flops, timing parameters | 5. Finite State Machines — Mealy/Moore models, state reduction, sequence detectors | 6. Memories and Programmable Logic — ROM/RAM, PLAs/PALs/FPGAs, design flow overview
Data Structures with C	1. Algorithm Analysis and Recurrence Basics — growth rates, recursion tree, master theorem (intuitive) | 2. Arrays, Linked Lists, and Iterators — singly/doubly/circular lists, operations, complexity | 3. Stacks and Queues — applications, infix-postfix conversion, circular and priority queues | 4. Trees and Balanced Trees — BST operations, AVL rotations, heaps and heapify | 5. Graphs and Traversals — adjacency models, BFS/DFS, connected components, topological sort | 6. Searching and Sorting Techniques — binary search, quicksort, mergesort, hashing fundamentals

Object-Oriented Programming (C++/Java)	1. OOP Principles and Class Design — abstraction, encapsulation, cohesion, UML basics | 2. Constructors, Destructors, and Overloading — operator/method overloading, copy control | 3. Inheritance and Polymorphism — virtual functions, interfaces, LSP, dynamic binding | 4. Exception Handling and Templates/Generics — robust error models, reusable components | 5. I/O Streams, Files, and Serialization — text/binary I/O, resource management | 6. Collections and Iterators — STL/Java Collections, maps/sets, iterators, algorithms library
Environmental Studies and Sustainability	1. Ecosystems and Biodiversity — structure, services, conservation strategies | 2. Natural Resources and Sustainable Use — water, forests, minerals, carrying capacity | 3. Pollution Control and Waste Management — air/water/soil noise, treatment technologies | 4. Environmental Impact Assessment — EIA process, mitigation, legal framework | 5. Climate Change and Energy — greenhouse effects, renewables, adaptation strategies | 6. Ethics, Equity, and Human Values — environmental governance, community engagement

Semester 3

Subject	Chapters & Topics (chapter names with key topics)
Discrete Mathematics for Computing	1. Logic and Propositional Calculus — predicates, inference, normal forms | 2. Proof Techniques and Induction — direct, contradiction, strong induction | 3. Sets, Relations, and Functions — equivalence, partial orders, composition | 4. Combinatorics and Counting — permutations, combinations, Pigeonhole, inclusion-exclusion | 5. Graph Theory Fundamentals — paths, cycles, trees, connectivity, planarity | 6. Algebraic Structures — semigroups, groups, rings (intro), applications in CS

Computer Organization and Architecture	1. Data Representation and Instruction Set Design — encoding, addressing modes | 2. CPU Datapath and Control — single-cycle/multi-cycle, microprogramming basics | 3. Pipelining and Hazards — structural/data/control hazards, forwarding | 4. Memory Hierarchy and Caches — locality, mapping, replacement, write policies | 5. I/O and Storage Systems — buses, interrupts, RAID, DMA, solid-state drives | 6. Performance and Parallelism — Amdahl’s law, SIMD/MIMD overview, benchmarks
Database Management Systems	1. Relational Model and SQL Foundations — schemas, constraints, basic queries | 2. Advanced SQL and Indexing — joins, subqueries, views, B+ trees | 3. Relational Algebra and Query Processing — operators, cost estimation, optimization | 4. Normalization and Schema Design — functional dependencies, 3NF/BCNF, anomalies | 5. Transaction Management and Concurrency — ACID, locking, isolation levels | 6. Recovery and NoSQL Overview — logging, checkpoints, key-value/document stores
Operating Systems	1. OS Structures and Processes — system calls, process model, context switching | 2. CPU Scheduling and Threads — FCFS/SJF/RR, multithreading, affinity | 3. Synchronization and Deadlocks — critical section, semaphores, deadlock handling | 4. Memory Management and Virtual Memory — paging, segmentation, replacement | 5. File Systems and Storage — allocation, directories, journaling, RAID | 6. Security and Protection — authentication, authorization, sandboxing, updates

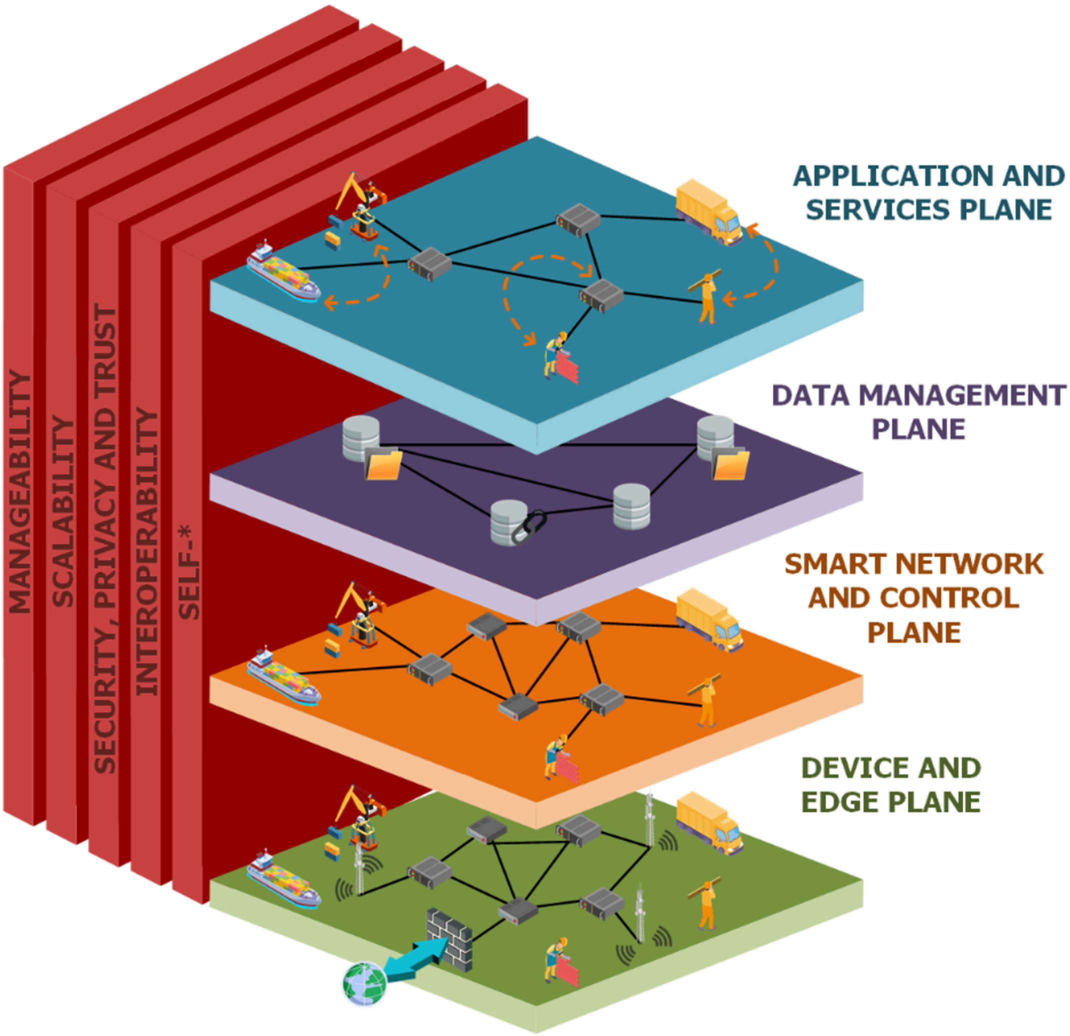
Design and Analysis of Algorithms	1. Algorithmic Strategies and Recurrences — divide-and-conquer, recurrence solving | 2. Greedy Algorithms and Matroids — activity selection, Huffman coding | 3. Dynamic Programming Paradigms — knapsack, LCS, matrix chain multiplication | 4. Graph Algorithms and Shortest Paths — Dijkstra, Bellman-Ford, MSTs | 5. NP-Completeness and Reductions — P vs NP, SAT, problem reductions | 6. Approximation and Randomized Methods — approximation ratios, randomization basics
Unix Utilities and Shell Programming	1. Linux Environment and File System — permissions, hierarchy, package tools | 2. Shell Scripting Basics — variables, control flow, functions, exit codes | 3. Text Processing Pipelines — grep, sed, awk, regular expressions | 4. Process and Job Control — ps/top, signals, background jobs, cron | 5. Version Control Fundamentals — Git workflows, branching, merging | 6. Build and Automation — makefiles, scripting for automation, logging

Semester 4

Subject	**Chapters & Topics (chapter names with key topics)**

Theory of Computation	1. Regular Languages and Finite Automata — DFAs, NFAs, regular expressions | 2. Properties and Limitations — pumping lemma, closure properties | 3. Context-Free Languages and PDA — grammars, derivations, pushdown automata | 4. Normal Forms and Parsing — CNF/GNF, parse trees, ambiguity | 5. Turing Machines and Computability — TM models, Church-Turing thesis | 6. Decidability and Undecidability — halting problem, reductions
Computer Networks	1. Network Architectures and Models — OSI/TCP-IP, layering principles | 2. Data Link and MAC Protocols — framing, error control, Ethernet, switching | 3. Network Layer and IP Routing — IPv4/IPv6, subnetting, OSPF/BGP | 4. Transport Protocols and Reliability — UDP, TCP flow and congestion control | 5. Application Layer Protocols — DNS, HTTP/HTTPS, email, CDN concepts | 6. Network Security Essentials — firewalls, TLS, VPNs, intrusion detection
Software Engineering	1. Software Process Models and Agile — waterfall, iterative, Scrum, Kanban | 2. Requirements Engineering — elicitation, specification, validation | 3. Software Design Principles — modularity, SOLID, design patterns overview | 4. Testing and Quality Assurance — unit/integration/system, coverage, CI | 5. Configuration and Release Management — versioning, branching, DevOps intro | 6. Project Management and Estimation — effort models, risk, metrics, documentation

Microprocessors and Interfacing	1. Microprocessor Architecture Basics — data paths, instruction cycles | 2. Assembly Language Programming — addressing modes, macros, procedures | 3. Memory and I/O Interfacing — mapped I/O, interrupts, DMA | 4. Peripherals and Timers — UART, SPI/I2C, timers/counters, ADC/DAC | 5. Microcontrollers Overview — 8051/ARM basics, GPIO, power modes | 6. Embedded System Case Studies — sensor interfacing, simple control loops
Web Technologies	1. HTML5 and CSS3 Foundations — semantic markup, responsive layouts, flex/grid | 2. JavaScript and DOM Programming — events, asynchronous calls, modules | 3. Front-End Framework Concepts — components, routing, state management | 4. Server-Side Programming — REST APIs, sessions, authentication | 5. Databases for the Web — schema design, ORMs, transactions, caching | 6. Web Security and Deployment — OWASP basics, HTTPS, containerized deployment
Numerical Methods and Scientific Computing	1. Floating-Point Arithmetic and Errors — conditioning, stability | 2. Root Finding and Nonlinear Equations — bisection, Newton-Raphson | 3. Linear Systems and Factorizations — Gaussian elimination, LU/QR | 4. Interpolation and Approximation — Newton/lagrange, least squares | 5. Numerical Differentiation and Integration — finite differences, quadrature | 6. ODE Initial Value Problems — Euler, Runge-Kutta, stiffness notions



Semester 5

Subject	Chapters & Topics (chapter names with key topics)

Compiler Design	1. Language Translators and Phases — front-end/back-end, passes, tools | 2. Lexical Analysis and Tokenization — regex, DFAs, scanner generators | 3. Syntax Analysis and Parsing — LL/LR, parser construction, error recovery | 4. Semantic Analysis and Type Checking — symbol tables, attribute grammars | 5. Intermediate Code and Optimization — three-address code, data-flow | 6. Code Generation and Runtime — register allocation, calling conventions
Artificial Intelligence	1. Intelligent Agents and Problem Formulation — environments, rationality | 2. Search Strategies and Heuristics — uninformed and informed search, A* | 3. Knowledge Representation and Reasoning — propositional/first-order logic | 4. Planning and Decision Making — classical planning, MDPs basics | 5. Uncertainty and Probabilistic Reasoning — Bayes nets, inference | 6. AI Applications and Ethics — NLP/vision overview, bias and fairness
Information Security and Cyber Laws	1. Security Principles and Threat Landscape — CIA triad, risk management | 2. Cryptography Foundations — symmetric/asymmetric, key management | 3. Network and System Security — firewalls, IDS/IPS, hardening | 4. Web and Application Security — SQLi, XSS, CSRF, secure coding | 5. Incident Response and Forensics — logging, triage, evidence handling | 6. Legal and Regulatory Framework — IT Act India, privacy, compliance

Distributed Systems	1. Architectural Models and Communication — client-server, RPC, sockets | 2. Synchronization and Coordination — clocks, mutual exclusion | 3. Consistency and Replication — consistency models, quorum, leases | 4. Fault Tolerance and Recovery — failures, checkpointing, Byzantine view | 5. Distributed File and Object Stores — NFS, GFS/HDFS, object storage | 6. Case Studies and Trends — microservices, containers, orchestration
Data Mining and Data Warehousing	1. Data Preprocessing and Feature Engineering — cleaning, transformation | 2. Association and Pattern Discovery — Apriori, FP-growth, interestingness | 3. Classification Techniques — decision trees, naive Bayes, evaluation | 4. Clustering Methods — k-means, hierarchical, DBSCAN | 5. Data Warehousing and OLAP — star/snowflake, cubes, ETL | 6. Big Data Platforms Overview — MapReduce, Spark, scalable mining
Professional Elective I (Example: Human-Computer Interaction)	1. HCI Foundations and User-Centered Design — goals, personas, scenarios | 2. Interaction Styles and Prototyping — low/high-fidelity, storyboards | 3. Cognitive Models and Usability — perception, mental models, heuristics | 4. Evaluation Methods and Metrics — usability tests, A/B, analytics | 5. Accessibility and Inclusive Design — WCAG, assistive technologies | 6. Design Systems and Patterns — components, consistency, design tokens

Semester 6

Subject	Chapters & Topics (chapter names with key topics)

Machine Learning	1. ML Foundations and Model Evaluation — bias-variance, cross-validation | 2. Linear Models and Regularization — ridge, lasso, logistic regression | 3. Tree-Based Methods and Ensembles — RF, gradient boosting, XGBoost | 4. Support Vector Machines and Kernels — margins, kernel trick | 5. Unsupervised Learning — PCA, k-means, anomaly detection | 6. Model Deployment and MLOps Basics — pipelines, monitoring, drift
Cloud Computing	1. Cloud Service Models and Architectures — IaaS/PaaS/SaaS, multi-cloud | 2. Virtualization and Containers — hypervisors, Docker fundamentals | 3. Cloud Storage and Databases — object/block/file, RDS/NoSQL | 4. Networking, Load Balancing, and CDN — VPC, gateways, traffic shaping | 5. Security, Identity, and Compliance — IAM, KMS, policies, audits | 6. Cloud-Native Design and Serverless — microservices, functions, event-driven
Internet of Things	1. IoT Architecture and Protocols — edge/fog/cloud, MQTT/CoAP | 2. Sensors, Actuators, and Embedded Nodes — selection, interfacing | 3. Connectivity and Gateways — BLE, Wi-Fi, LPWAN, IoT hubs | 4. Data Ingestion and Processing — stream processing, time-series | 5. Security and Privacy in IoT — device identity, secure updates | 6. Applications and Case Studies — smart home, industry, healthcare

Mobile Application Development	1. Mobile Platforms and UI Paradigms — Android/iOS, adaptive layouts | 2. Activity/Fragment Lifecycles and Navigation — state handling | 3. Data Storage and Networking — SQLite/Room, REST/GraphQL | 4. Reactive Programming and MV* Patterns — LiveData/Rx/Compose | 5. Performance, Battery, and Accessibility — profiling, a11y | 6. Testing and Publishing — unit/UI tests, signing, store release
Big Data Analytics	1. Big Data Ecosystem and HDFS — characteristics, architecture | 2. MapReduce and YARN — execution model, optimizations | 3. Apache Spark Core and SQL — RDDs, DataFrames, caching | 4. Streaming and Event Processing — Spark Streaming, Kafka | 5. NoSQL Systems and Column Stores — HBase/Cassandra, modeling | 6. Scalable ML Pipelines — Spark MLlib, feature engineering
Professional Elective II (Example: Blockchain Technologies)	1. Cryptographic Primitives for Blockchain — hashes, signatures | 2. Distributed Ledgers and Consensus — PoW, PoS, BFT variants | 3. Smart Contracts and Platforms — EVM, Solidity, safety | 4. Blockchain Security and Privacy — attacks, mixers, ZKPs | 5. Enterprise and Permissioned Chains — Hyperledger, governance | 6. Use Cases and Limitations — finance, supply chain, scalability

Semester 7

Subject	**Chapters & Topics (chapter names with key topics)**

Deep Learning	1. Neural Network Foundations — perceptrons, backpropagation | 2. Convolutional Networks for Vision — architectures, regularization | 3. Sequence Models and RNNs — LSTM/GRU, attention basics | 4. Optimization and Generalization — SGD variants, normalization | 5. Advanced Topics and Transformers — self-attention, transfer learning | 6. Engineering DL Systems — mixed precision, serving, hardware accelerators
Natural Language Processing	1. Text Processing and Representations — tokenization, embeddings | 2. Classical NLP Models — n-grams, HMMs, CRFs | 3. Neural NLP and Sequence-to-Sequence — encoder-decoder | 4. Contextual Language Models — BERT family, fine-tuning | 5. Information Extraction and QA — NER, relation extraction | 6. Evaluation, Bias, and Ethics — metrics, fairness, safety
Software Project Management and Agile	1. Project Initiation and Scope — charters, stakeholder mapping | 2. Estimation and Planning — story points, velocity, release plans | 3. Agile Execution and DevOps — CI/CD, trunk-based development | 4. Risk and Quality Management — risk registers, SQA practices | 5. Metrics and Governance — DORA metrics, dashboards, audits | 6. Contracts and Procurement — vendor management, SLAs

High-Performance Computing (Parallel and GPU)	1. Parallel Architectures and Models — shared/distributed memory | 2. Parallel Programming with OpenMP — tasks, synchronization | 3. Distributed Programming with MPI — communicators, collectives | 4. GPU Computing with CUDA — threads, memory hierarchy | 5. Performance Analysis and Tuning — profiling, scaling | 6. Parallel Algorithms and Case Studies — stencil, reductions
Professional Elective III (Example: Computer Vision)	1. Image Formation and Camera Models — projections, calibration | 2. Image Processing and Features — filtering, SIFT/SURF | 3. Recognition and Detection — HOG, SVMs, cascades | 4. Deep Vision Architectures — CNNs, detection/segmentation | 5. 3D Vision and Geometry — stereo, structure from motion | 6. Applications and Evaluation — tracking, metrics, datasets
Open Elective I (Example: Augmented and Virtual Reality)	1. Visual Computing Foundations — rendering pipeline, shaders | 2. Tracking and Pose Estimation — IMU/SLAM, sensor fusion | 3. Interaction and UX for XR — input models, comfort | 4. AR Frameworks and Toolchains — ARCore/ARKit, Unity | 5. VR Systems and Content — locomotion, presence, nausea | 6. Deployment and Ethics — safety, privacy, accessibility

Semester 8

Subject	Chapters & Topics (chapter names with key topics)

DevOps and MLOps	1. Continuous Integration and Delivery — pipelines, artifacts, environments | 2. Infrastructure as Code and Containers — Terraform, Kubernetes | 3. Observability and Reliability — logging, metrics, SLOs, incident response | 4. Security and Compliance in CI/CD — secrets, policies, SBOMs | 5. MLOps Lifecycle and Model Management — feature stores, registries | 6. Serving, Monitoring, and Governance — A/B, canary, drift, lineage
Entrepreneurship, Innovation, and IPR	1. Opportunity Discovery and Validation — problem-solution fit, JTBD | 2. Business Models and Strategy — canvases, moats, unit economics | 3. Product Management and Roadmaps — MVPs, prioritization, feedback loops | 4. Financing and Growth — bootstrapping, venture funding, metrics | 5. Intellectual Property and Legal — patents, trademarks, licensing | 6. Tech Venture Case Studies — successes, failures, ethical considerations
Professional Ethics and Social Responsibility	1. Ethical Theories and Decision Frameworks — deontological, consequentialist | 2. Data Privacy and Governance — consent, anonymization, compliance | 3. Responsible AI and Fairness — bias detection, explainability | 4. Cyber Ethics and Digital Rights — surveillance, free speech, inclusion | 5. Sustainability in Technology — e-waste, green computing | 6. Professional Conduct and Leadership — codes, accountability

Open Elective II (Example: Edge Computing)	1. Edge Architectures and Orchestration — MEC, K3s, function offloading | 2. Edge AI and Acceleration — model compression, NN inference | 3. Scheduling and Resource Management — latency-aware placement | 4. Data Management at the Edge — stream processing, privacy | 5. Security and Trust at the Edge — attestation, secure enclaves | 6. Use Cases and Benchmarks — industrial IoT, autonomous systems
Capstone Project (Guidelines and Milestones)	1. Problem Definition and Literature Survey — scope, novelty, prior art | 2. Requirements and System Design — architecture, interfaces, risks | 3. Implementation Planning and Iterations — sprints, CI, coding standards | 4. Verification, Validation, and Testing — unit/system/acceptance tests | 5. Documentation and Presentation — reports, demos, poster, viva | 6. Deployment and Impact Assessment — metrics, user feedback, future work

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✅ Tip: Map the chapter names to your university's official syllabus code. You can split or merge chapters as per weekly lecture hours and add lab components beneath each subject for a complete teaching plan.

If you want, I can tailor this to a specific university or replace the elective examples with your chosen elective lists and expand lab syllabi into weekly experiments.